

## LINZ STADT, Austria

WMO: 110600

Lat: 48.296N

Lon: 14.285E

Elev: 262

StdP: 98.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.1       | -7.0 | -12.5                         | 1.3 | -8.0 | -10.0 | 1.6 | -5.7 | 3.5                   | 1.7  | 3.4 | 1.7  | 1.3                   | 220  | 0.419 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.3                    | 31.3            | 20.9 | 29.3 | 20.2 | 27.7 | 19.4 | 21.9                | 29.0 | 20.9 | 27.7 | 20.1 | 26.1 | 1.5                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.5                            | 14.7 | 24.9 | 18.7 | 13.9 | 23.6 | 17.9 | 13.3 | 22.8 | 65.0          | 29.0 | 61.5 | 27.9 | 58.6 | 26.2 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 3.5           | 3.2  | 2.7 | -11.1                      | 34.1 | 3.3                | 1.5 | -13.5                                              | 35.2 | -15.4      | 36.1 | -17.3      | 36.9 | -19.7      | 38.1 |     |     |
| (5)               |      |     | -11.5                      | 23.4 | 3.2                | 1.0 | -13.8                                              | 24.1 | -15.7      | 24.8 | -17.5      | 25.3 | -19.9      | 26.1 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.7        | -0.1  | 2.1   | 6.1   | 11.5  | 15.6  | 18.9  | 20.5  | 20.3  | 15.5  | 10.6  | 5.5   | 1.5   | (6)  |  |
|                                                                        | DBStd       | 8.13        | 3.90  | 4.02  | 3.95  | 4.25  | 3.74  | 3.82  | 3.35  | 3.38  | 3.28  | 3.54  | 3.51  | 3.63  | (7)  |  |
|                                                                        | HDD10.0     | 1143        | 313   | 222   | 129   | 33    | 2     | 0     | 0     | 0     | 1     | 37    | 141   | 265   | (8)  |  |
|                                                                        | HDD18.3     | 3033        | 572   | 454   | 378   | 207   | 101   | 39    | 17    | 18    | 97    | 241   | 386   | 523   | (9)  |  |
|                                                                        | CDD10.0     | 1403        | 0     | 1     | 9     | 78    | 176   | 268   | 327   | 319   | 166   | 54    | 5     | 0     | (10) |  |
|                                                                        | CDD18.3     | 252         | 0     | 0     | 0     | 2     | 17    | 56    | 86    | 79    | 12    | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1999        | 0     | 0     | 0     | 27    | 140   | 467   | 683   | 614   | 67    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 587         | 0     | 0     | 0     | 3     | 22    | 138   | 221   | 194   | 9     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 1.2         | 1.2   | 1.3   | 1.3   | 1.4   | 1.4   | 1.3   | 1.2   | 1.2   | 1.1   | 1.1   | 1.1   | 1.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 840         | 39    | 54    | 55    | 62    | 96    | 116   | 96    | 90    | 61    | 52    | 60    | 60    | (15) |  |
|                                                                        | PrecMax     | 1081        | 85    | 192   | 109   | 100   | 170   | 171   | 190   | 187   | 95    | 140   | 96    | 158   | (16) |  |
|                                                                        | PrecMin     | 535         | 4     | 14    | 14    | 20    | 43    | 52    | 35    | 20    | 27    | 3     | 25    | 6     | (17) |  |
|                                                                        | PrecStd     | 160         | 28    | 44    | 27    | 25    | 44    | 34    | 45    | 47    | 21    | 38    | 21    | 47    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 11.7  | 13.8  | 19.5  | 26.4  | 29.0  | 32.5  | 33.6  | 33.6  | 27.9  | 22.3  | 17.0  | 11.2  | (19) |  |
|                                                                        |             | MCWB        | 8.5   | 8.7   | 11.5  | 15.7  | 19.1  | 21.1  | 21.5  | 20.9  | 19.3  | 15.4  | 12.1  | 8.9   | (20) |  |
|                                                                        | 2%          | DB          | 8.7   | 11.2  | 16.7  | 23.3  | 26.9  | 30.2  | 31.1  | 30.9  | 25.3  | 19.3  | 13.9  | 9.0   | (21) |  |
|                                                                        |             | MCWB        | 6.5   | 7.6   | 10.5  | 14.7  | 18.2  | 20.8  | 20.9  | 20.7  | 18.2  | 14.5  | 10.8  | 7.0   | (22) |  |
|                                                                        | 5%          | DB          | 6.7   | 9.3   | 14.2  | 21.1  | 24.8  | 28.1  | 29.1  | 28.7  | 23.1  | 17.5  | 11.9  | 7.3   | (23) |  |
|                                                                        |             | MCWB        | 5.1   | 6.4   | 9.2   | 13.6  | 17.1  | 19.7  | 20.3  | 20.0  | 17.3  | 13.8  | 9.7   | 5.8   | (24) |  |
|                                                                        | 10%         | DB          | 5.0   | 7.4   | 12.1  | 18.9  | 22.7  | 26.1  | 27.2  | 26.8  | 21.1  | 15.7  | 10.1  | 6.0   | (25) |  |
|                                                                        |             | MCWB        | 3.7   | 5.2   | 8.1   | 12.5  | 15.7  | 18.8  | 19.4  | 19.2  | 16.2  | 13.0  | 8.4   | 4.9   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.0   | 9.7   | 12.3  | 16.4  | 20.3  | 22.9  | 23.2  | 22.8  | 20.0  | 16.6  | 12.7  | 9.1   | (27) |  |
|                                                                        |             | MCDB        | 11.0  | 12.9  | 18.0  | 24.7  | 26.9  | 30.1  | 30.8  | 30.5  | 25.3  | 19.9  | 15.9  | 10.7  | (28) |  |
|                                                                        | 2%          | WB          | 6.8   | 8.0   | 10.9  | 15.1  | 18.7  | 21.5  | 21.9  | 21.4  | 18.8  | 15.3  | 11.2  | 7.5   | (29) |  |
|                                                                        |             | MCDB        | 8.4   | 10.9  | 15.6  | 22.5  | 25.4  | 28.4  | 28.9  | 28.8  | 24.0  | 18.0  | 13.5  | 8.7   | (30) |  |
|                                                                        | 5%          | WB          | 5.3   | 6.6   | 9.7   | 13.9  | 17.6  | 20.4  | 20.9  | 20.5  | 17.8  | 14.3  | 9.8   | 6.0   | (31) |  |
|                                                                        |             | MCDB        | 6.6   | 8.9   | 13.4  | 20.0  | 23.7  | 26.6  | 27.5  | 27.2  | 22.2  | 16.8  | 11.8  | 7.0   | (32) |  |
|                                                                        | 10%         | WB          | 3.8   | 5.4   | 8.6   | 12.8  | 16.5  | 19.4  | 20.0  | 19.7  | 16.9  | 13.2  | 8.6   | 4.9   | (33) |  |
|                                                                        |             | MCDB        | 4.9   | 7.2   | 11.6  | 18.2  | 21.7  | 24.9  | 25.8  | 25.6  | 20.3  | 15.4  | 9.9   | 6.0   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.0   | 5.8   | 7.7   | 9.2   | 9.0   | 9.0   | 9.3   | 9.2   | 7.9   | 6.8   | 4.5   | 3.8   | (35) |  |
|                                                                        |             | MCDBR       | 5.9   | 9.4   | 12.1  | 13.2  | 12.8  | 12.6  | 12.6  | 12.7  | 11.1  | 9.4   | 7.8   | 6.0   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 4.2   | 6.0   | 6.6   | 6.2   | 5.7   | 5.1   | 4.7   | 4.6   | 4.8   | 5.2   | 4.9   | 4.6   | (37) |  |
|                                                                        |             | MCWBR       | 5.8   | 8.7   | 11.1  | 12.3  | 11.8  | 11.6  | 11.3  | 11.5  | 9.9   | 8.5   | 7.3   | 5.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.300 | 0.326 | 0.367 | 0.403 | 0.411 | 0.414 | 0.422 | 0.416 | 0.388 | 0.356 | 0.329 | 0.300 | (40) |  |
|                                                                        |             | taud        | 2.387 | 2.339 | 2.294 | 2.250 | 2.270 | 2.294 | 2.275 | 2.307 | 2.357 | 2.429 | 2.455 | 2.438 | (41) |  |
|                                                                        | Ebn at Noon |             | 768   | 824   | 842   | 846   | 852   | 850   | 836   | 824   | 812   | 782   | 729   | 724   | (42) |  |
|                                                                        |             | Edh at Noon | 68    | 89    | 109   | 127   | 129   | 127   | 128   | 118   | 102   | 81    | 64    | 58    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.04  | 1.94  | 2.99  | 4.27  | 4.95  | 5.41  | 5.27  | 4.68  | 3.35  | 2.13  | 1.11  | 0.82  | (44) |  |
|                                                                        | RadStd      |             | 0.16  | 0.25  | 0.37  | 0.52  | 0.58  | 0.47  | 0.50  | 0.49  | 0.43  | 0.23  | 0.13  | 0.10  | (45) |  |

## Historical Trends

|                               |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                               |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                               |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only             |  | +0.66 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (104 neighbors) |  | +0.52 | N/A     | N/A    | N/A     | +0.39 | +0.34 | N/A         | -141    | +70     | +31     |

Nomenclature: See separate page